

# MOHAMMED FAIHAN

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Portfolio: [md-fn.github.io](https://md-fn.github.io) | [LinkedIn](#)

Creative engineer and maker with expertise in robotics, embedded systems, and 3D design. Problem solver with a passion for bringing ideas to life through hardware prototyping, CAD design, and digital fabrication. Experienced in designing systems with hardware & software and creating digital creative content.

## EDUCATION

**BSc in Computer Engineering**

**Ajman University**, Ajman, UAE  
Expected Graduation: May 2027

**High School**

**Habitat School**, Ajman, UAE  
Graduated 2023

## PROFESSIONAL EXPERIENCE

### Freelance Creative Designer & 3D Specialist

2024 – Present

- Deliver architectural visualizations and interior design renderings for commercial clients using Blender.
- Produce high-quality 3D CAD designs for manufacturing in Autodesk Fusion

### Content Creator – Robotics Program

**Inventify**, Dubai, UAE | Summer 2024 (Part-time)

- Planned, filmed, and edited robotics course content for educational platform for kids aged 9 -15
- Designed and 3D-printed custom mechanical components for robotics kits and solved problems across work using my skills.

## PROJECTS

### Mechanical Catapult – IEEE UAE Student Day- 2025: Engineering Design Project, 2nd Place

- Designed and constructed precision catapult mechanism to accurately launch projectiles at variable distances; tested and optimized mechanical linkages for consistency

### Color sorting with a 3DOF Robotic arm with PIC18 – Embedded Systems Course Project

- Designed a 3DOF mechanical arm in Fusion with complex linkages and joints, then 3D printed and assembled it.
- Integrated it with a TCS3200 Color Sensor, Stepper motors and a Servo, so the robot could accurately sort colored objects into its respective bins.

### **Custom Core-XY 3D Printer**

- Designed and engineered printer components in Autodesk Fusion 360; optimized for manufacturability and assembly
- Constructed printer from aluminum extrusions, 3D-printed parts, and salvaged components from an old 3d Printer; assembled and calibrated mechanical system
- Configured Klipper firmware and optimized print profiles for high-quality output

### **Smart Pipeline Monitoring Robot – DEWA AI Championship Winner 2022**

- Designed and built an autonomous robot for oil and gas pipeline inspection using Computer Vision
- Integrated mechanical movement system with onboard camera, Lidar, environmental sensors, and microcontroller-based control (ESP32)

**Additional projects updated in portfolio:** [md-fn.github.io](https://md-fn.github.io)

## **TECHNICAL SKILLS**

**Robotics, Electronics & Hardware:** Embedded Systems, Arduino & ESP32, Raspberry Pi, circuit design & prototyping, 3D printing, mechanical prototyping, sensor integration.

**3D Printing:** Operating 5+ different 3D printers, Repairing and Modifying 3D printers.

**3D CAD & Design:** Autodesk Fusion 360, mechanical design, parametric modeling, 3D printer slicing.

**3D Rendering & Animation:** Blender (modeling, rendering, animation, visualization)

**Graphic Design & Video Editing:** Affinity Suite, Inkscape, DaVinci Resolve (video editing).

**Programming Languages:** C, C++, Python,

## **CERTIFICATIONS & ACHIEVEMENTS**

- IEEE UAE Engineering Design Project – 2nd Place (2025)
- Object-Oriented C++ Specialization – Coursera (Inheritance, Encapsulation; Pointers, Arrays, Recursion)
- Duke University & Codio – C++ Programming Certificates

## **INTERESTS & HOBBIES**

Robotics, 3D Printing, Computer Hardware, Design, Fitness, Photography, Cinematography, Mechanical Engineering.